

UDP device node data

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UDP device node data

Node data type

- UPLINK
- DOWNLINK (Time calibration package)
- DOWNLINK (Control command)
- UPLINK (Command execution result)

Device node data interaction process

The data is in the format of byte stream in the process of SOCKET interaction, but the actual format of device data is HEX string. You need to convert them to each other, that is, the byte stream format data is converted to HEX string format.

1. The device node will actively connect to the base station by manually triggering activation or timing activation. If the connection is successful, it will obtain a temporary IP address and send UDP_UPLINK information to the IP address server set by the program through random ports.
2. Send UDP_UPLINK information to the IP address server set by the program using a random port from the assigned temporary IP address
 - a. If the connection fails or does not receive the school, it will be resent, the default number of retransmissions is 2, and the next reporting interval will be random.
 - b. When there is no data within 20s, the device will disconnect the connection (that is, the time difference between sending the downlink command and the uplink data to the server must not exceed 20s, otherwise it will be invalid).
3. Generate time correction commands to be sent.
 - a. IP address that obtains the source IP address of uplink data based on socket interaction and saves it for downlink transmission.
It is only valid for the current process.
 - b. The server needs to parse the corresponding SN number according to the uplink data, assemble the timing command according to the SN number, and send it.
4. According to the IP,PORT obtained in the third step, using SOCKET interaction, the downlink command in HEX string format is converted into a byte stream to be sent.

5. Assemble downlink control commands to be sent.
 - a. Downlink control command assembly reference.
 - b. Only one control command can be sent at a time, and execution may fail if there are more than one.
6. According to the IP,PORT obtained in the third step, using SOCKET interaction, the downlink command in HEX string format is converted into a byte stream to be sent.
7. After the command is executed successfully, the response data is replied, according to the message type(2).
 - a. The status after execution is returned. The command is executed successfully, and some parameters do not support setting. You need to determine whether the set parameters are valid according to the returned status.
 - b. There will be no reply for unsupported commands (there will be conflicts when multiple commands are received at the same time), and the process will end.
8. If there are still downlink control commands to be sent, proceed to step 6, and if not, end the current process.

figure 1: udp

UPLINK

```
01010002cfbd3c00bcff86ab62626e642f332f3002693132333435363738390d1a64351eb70e6555
54432b3801634e426807190168110112655056332e30136c56332e30305f32313031303414001700
a662626e652f38302f3000664c58432d323001020600101a0098967f151a64351eb7a462626e652f
38312f30030101000201a362626e652f38322f3000000100a262626e652f38342f30001a00015180
a562626e652f39392f30016f3836373732343033313736383430380b39040f0d39035b0e381fda50
```

```
0101      :1.01 Fixed value
00        :Message type,Fixed value ,0:UPLINK
02        :Function code,Fixed value
cfbd      :Message ID, Random at 1-65535
3c        :Message format, 60-CBOR
00bc      :Data field length
ff        :Delimiter FF
86ab62626e642f332f3002693132333435363738390d1a64351eb70e655554432b3801634e426807
190168110112655056332e30136c56332e30305f32313031303414001700a662626e652f38302f30
00664c58432d323001020600101a0098967f151a64351eb7a462626e652f38312f30030101000201
a362626e652f38322f3000000100a262626e652f38342f30001a00015180a562626e652f39392f30
016f3836373732343033313736383430380b39040f0d39035b0e381f: Data domain data--CBOR
da50      :CRC16
```

CBOR--DATA

The cbor protocol is a common encoding protocol, you can also refer to this page to see the structure of the data.

CBOR: <http://cbor.me/>

Data domain data

```
86ab62626e642f332f3002693132333435363738390d1a64351eb70e655554432b3801634e426807
190168110112655056332e30136c56332e30305f32313031303414001700a662626e652f38302f30
00664c58432d323001020600101a0098967f151a64351eb7a462626e652f38312f30030101000201
a362626e652f38322f3000000100a262626e652f38342f30001a00015180a562626e652f39392f30
016f3836373732343033313736383430380b39040f0d39035b0e381f
```

```
[{"bn": "/3/0", 2: "123456789", 13: 1681202871, 14: "UTC+8", 1: "NBh", 7: 360,
17: 1, 18: "PV3.0", 19: "v3.00_210104", 20: 0, 23: 0}, {"bn": "/80/0", 0: "LXC-
20", 1: 2, 6: 0, 16: 9999999, 21: 1681202871}, {"bn": "/81/0", 3: 1, 1: 0, 2: 1},
{"bn": "/82/0", 0: 0, 1: 0}, {"bn": "/84/0", 0: 86400}, {"bn": "/99/0", 1:
"867724031768408", 11: -1040, 13: -860, 14: -32}]
```

DOWNLINK (Time calibration package)

The calibration time of the device is used to indicate that the device has successfully interacted with the server and to calibrate the time of the server, so it needs to be sent every time it is reported.

The command is a response, that is, there will be no response, regardless of whether it is executed successfully or not

```
01010245fb6e3c0025ff81a502693132333435363738390d1a6965eb8e0e655554432b3818190162
626e642f332f30a487
```

```
0101      :1.01 Fixed value
02        :Message type, Fixed value,If the time calibration command is used,
there will be no data reply.
45        :Function code: 69, Fixed value
fb6e      :Message ID:64366, Random at 1-65535
3c        :Message format, 60-CBOR
0025      :Data field length: 37 bytes
ff        :Delimiter FF
81a502693132333435363738390d1a6965eb8e0e655554432b3818190162626e642f332f30: Data
domain data--CBOR
a487      :CRC16
```

Data domain data

The calibration needs to be sent according to the SN value reported by the device, otherwise the command device will not execute

```
81a502693132333435363738390d1a6965eb8e0e655554432b3818190162626e642f332f30
```

```
[{2: "123456789", 13: 1768287118, 14: "UTC+8", 25: 1, "bn": "/3/0"}]
```

```
Object: /3/0    //Data objects, device information
2: 123456789    // serial number( meter identification number )
14: UTC+8      // UTC+8
3: 1768287118  // UTC+8 2026-01-13 14:51:58
25: 1          // follow up data: None
```

DOWNLINK (Control command)

Downlink control command, which only sends a single parameter control information at a time, and may fail when there are multiple parameters.
The command uses a confirmation frame, and if the execution is successful, the successful reply will be reported.

```
01010003dd6e3c0030ff82a3001a004f1a00166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f30e964
```

```
0101      :1.01 Fixed value
00        :Message type,Fixed value
03        :Function code,Fixed value
dd6e      :Message ID, Random at 1-65535
3c        :Message format, 60-CBOR
0030      :Data field length
ff        :Delimiter FF
82a3001a004f1a00166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f30: Data domain data--CBOR
e964      :CRC16
```

Data domain data

```
82a3001a004f1a00166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f30
```

```
[{0: 5184000, 22: "867724031768408", "bn": "/84/0"}, {2: 2018, "bn": "/70/0"}]
```

0: 5184000: **0**:Reporting interval, **5184000**:5184000s

22: "867724031768408": **22**:IMEI specifies the serial number, **867724031768408**:IMEI

"bn": "/84/0": Bn value, refer to the content of the agreement

{2: 2018, "bn": "/70/0"}: Fill in with fixed value

Valve control

Valve control command

bn:/81/0

key: 0

- Open valve

```
01010003c4a23c002cff82a30000166f38363737323430333137363834303862626e652f38312f30a2021907e262626e652f37302f301072
```

```
82a30000166f38363737323430333137363834303862626e652f38312f30a2021907e262626e652f37302f30
```

```
[{0: 0, 22: "867724031768408", "bn": "/81/0"}, {2: 2018, "bn": "/70/0"}]
```

- Close valve

```
0101000351d53c002cff82a30001166f38363737323430333137363834303862626e652f38312f30a2021907e262626e652f37302f3049fb
```

```
82a30001166f38363737323430333137363834303862626e652f38312f30a2021907e262626e652f37302f30
```

```
[{0: 1, 22: "867724031768408", "bn": "/81/0"}, {2: 2018, "bn": "/70/0"}]
```

Calibration device node reading

The node reading of the calibration device is 888L

bn:/80/0

key: 16

```
01010003773e3c002eff82a310190378166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f30f492
```

```
82a310190378166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f30
```

```
[{16: 888, 22: "867724031768408", "bn": "/80/0"}, {2: 2018, "bn": "/70/0"}]
```

Reporting cycle interval

Set the interval of the reporting week to 86400s

bn:/84/0

key: 0

```
0101000305c33c0030ff82a3001a00015180166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f3030c3
```

```
82a3001a00015180166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f30
```

```
[{0: 86400, 22: "867724031768408", "bn": "/84/0"}, {2: 2018, "bn": "/70/0"}]
```

Manual time calibration

Calibration device time is timestamp 1682524800

bn:/3/0

key: 13

```
010100038a143c002fff82a30d1a64494a80166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f309685
```

```
82a30d1a64494a80166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f30
```

```
[{13: 1682524800, 22: "867724031768408", "bn": "/3/0"}, {2: 2018, "bn": "/70/0"}]
```

TimeZone

Set time zone to UTC+2

bn:/3/0

key: 14

```
010100033dcf3c0030ff82a30e655554432b32166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f302ace
```

```
82a30e655554432b32166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f30
```

```
[{"14": "UTC+2", "22": "867724031768408", "bn": "/3/0"}, {"2": 2018, "bn": "/70/0"}]
```

SN

Set the SN number of the device to 666666

bn:/3/0

key: 2

```
01010003116c3c0031ff82a302663636363636166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f30ec71
```

```
82a30266363636363636166f38363737323430333137363834303862626e642f332f30a2021907e262626e652f37302f30
```

```
[{"2": "666666", "22": "867724031768408", "bn": "/3/0"}, {"2": 2018, "bn": "/70/0"}]
```

IP

IP: 47.92.222.233 :1822

bn:/99/0

key: 4

```
0101000366ad3c003eff82a3047234372e39322e3232322e32333333a31383232166f38363737323430333137363834303862626e652f39392f30a2021907e262626e652f37302f30edd8
```

```
82a3047234372e39322e3232322e32333333a31383232166f38363737323430333137363834303862626e652f39392f30a2021907e262626e652f37302f30
```

```
[{"4": "47.92.222.233:1822", "22": "867724031768408", "bn": "/99/0"}, {"2": 2018, "bn": "/70/0"}]
```

PN

Set PN value to 1L/P

bn:/80/0

key: 2

```
01010003ffbf3c002cff82a30201166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f306d06
```

```
82a30201166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f30
```

```
[{2: 1, 22: "867724031768408", "bn": "/80/0"}, {2: 2018, "bn": "/70/0"}]
```

Maximum meter reading

Set the maximum reading of the device to: 999999999

bn:/80/0

key: 7

```
0101000306c53c0030ff82a3071a3b9ac9ff166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f30e613
```

```
82a3071a3b9ac9ff166f38363737323430333137363834303862626e652f38302f30a2021907e262626e652f37302f30
```

```
[{7: 999999999, 22: "867724031768408", "bn": "/80/0"}, {2: 2018, "bn": "/70/0"}]
```

Reporting range

Set the time range for reporting by the device, which is valid only when the reporting cycle interval is ≤ 86400 s.

Set the discrete reporting interval to 0-6: 00.

bn:/84/0

key: 5

```
010100038dac3c002cff82a30514166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f300a31
```

```
82a30514166f38363737323430333137363834303862626e652f38342f30a2021907e262626e652f37302f30
```

```
[{5: 20, 22: "867724031768408", "bn": "/84/0"}, {2: 2018, "bn": "/70/0"}]
```

20: 0020 0--20/2 ,It means to report randomly in the range of 0-10:00

321: 0321 3--3+21/2 ,Means to report randomly in the range of 3-13:30.

236: 0236 2--2+36/2 ,It means to report randomly in the range of 2:00-20:00.

Read dense data

read dense data

Get the monthly frozen data of the device: **2309** Monthly freeze data for /9/23

Get the daily freeze data of the device: **230911** Daily Freeze Data for 11/9/23

bn:/80/0

key: 28

```
0101000347663c0031ff82a3166f383637373234303331373638343038181c1a0002e8fe62626e652f38302f30a2021907e262626e652f37302f305e2a
```

```
82a3166f383637373234303331373638343038181c1a0002e8fe62626e652f38302f30a2021907e262626e652f37302f30
```

```
[{28: 231011, 22: '867724031768408', 'bn': '/80/0'}, {2: 2018, 'bn': '/70/0'}]
```

UPLINK (Command execution result)

It is usually the data reported by the device to the server, which is used for the parameter values returned after successful execution.

The command is reported using the response value.

```
01010244dd6e3c005dff84a262626e652f37302f30021907e2a762626e652f38302f3000664c58432d32300102020206001000151a64351ad2a262626e652f38342f300000a462626e652f39392f30016f3836373732343033313736383430380d3902a80e381d48e9
```

```
0101      :1.01 Fixed value
02        :Message type,Fixed value ,the response value
44        :Function code,Fixed value
dd6e      :Message ID, Random at 1-65535
3c        :Message format, 60-CBOR
005d      :Data field length
ff        :Delimiter FF
84a262626e652f37302f30021907e2a762626e652f38302f3000664c58432d32300102020206001000151a64351ad2a262626e652f38342f300000a462626e652f39392f30016f3836373732343033313736383430380d3902a80e381d: Data domain data--CBOR
48e9      :CRC16
```

Data domain data

```
84a262626e652f37302f30021907e2a762626e652f38302f3000664c58432d32300102020206001000151a64351ad2a262626e652f38342f300000a462626e652f39392f30016f3836373732343033313736383430380d3902a80e381d
```

```
[{"bn": "/70/0", 2: 2018}, {"bn": "/80/0", 0: "Lxc-20", 1: 2, 2: 2, 6: 0, 16: 0, 21: 1681201874}, {"bn": "/84/0", 0: 0}, {"bn": "/99/0", 1: "867724031768408", 13: -681, 14: -30}]
```